

pruritus in psoriasis has not been effectively studied. The lack of correlation between pruritus severity and psoriasis disease severity that we have observed, coupled with the high prevalence of pruritus and unmet need, warrants greater attention to pruritus by clinicians and researchers to develop targeted antipruritic therapies for patients with psoriasis and other dermatologic conditions.

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Treatment of erythropoietic protoporphyria with the oral sorbent colestipol: A proof-of-concept clinical trial

To the Editor: Erythropoietic protoporphyria (EPP) is characterized by cutaneous photosensitivity and, rarely, potentially lethal liver dysfunction. The accumulation of free protoporphyrin IX in tissues and organs of patients with EPP results from mutations in genes coding for enzymes involved in heme biosynthesis.¹ Hepatocyte protoporphyrin accumulation, the cause of hepatic failure, is augmented by both its transfer from blood and the enterohepatic recirculation of porphyrins after they leave the biliary tract.² Several treatments, often marginally effective, are available for the cutaneous manifestations.³ No medical therapies prevent the progression of hepatic dysfunction.

In vitro, cholestyramine and activated charcoals can irreversibly bind porphyrins, including protoporphyrin.⁴ Their irreversible binding of ileum-based protoporphyrin, preventing its enterohepatic reabsorption, may lead to decreased accumulation of protoporphyrin and be therapeutic in both skin and liver. This study addresses the hypothesis that a useful sorbent, colestipol, effectively diminishes the cutaneous and hepatic manifestations of EPP.

This study, approved by Partners HealthCare Investigational Review Board, included 3 adult women with EPP and cutaneous but no hepatic manifestations. After a period of no therapy (during late winter and spring), each subject received oral colestipol at 2 g twice daily (during summer and early autumn). In both phases, subjects used a stopwatch to determine the time to the onset of discomfort with sun exposure and completed a standardized questionnaire, addressing the affected areas, and the nature and severity of manifestations. Blood samples were analyzed for liver function and

Table I. Sun tolerance with colestipol 2 g twice daily

Days of treatment	Onset of sensitivity to sun*					
	Subject 1		Subject 2		Subject 3	
	Min	Fold ↑†	Min	Fold ↑†	Min	Fold ↑†
0	8	—	1.6	—	9.3	—
60-65	74	9.3	3.3	2.1	120	13
125-127	190	24	2.5	1.6	∞‡	∞‡
157-163	220	28	7.1	4.4	∞‡	∞‡

*Determined by each subject recording the onset of skin discomfort with a stopwatch.

†Colestipol-mediated increase in sun tolerance relative to baseline tolerance. The hazard per day ratio using the Cox proportional hazards model = 0.96; *P* = .018.

‡No development of skin sensitivity.

Table II. Erythrocyte (red blood cell) and plasma protoporphyrin concentrations before and during colestipol treatment 2 g twice daily

Mean free protoporphyrin concentration, $\mu\text{g/dL}$						
Subject	Pretreatment			During treatment		
	N*	RBC	Plasma	N*	RBC	Plasma
1	2	2882	3.4	3	2252	2.6
2	2	1259	20.5	3	1431	13.5
3	2	1660	1.6	3	1453	1.3
Mean $\pm$ SD		1934 $\pm$ 714	9.9 $\pm$ 13.4		1712 $\pm$ 455	5.8 $\pm$ 8.3

SD, Standard deviation; RBC, red blood cell.

*Number of samples from each subject analyzed for erythrocyte and plasma-free protoporphyrin before or during treatment. The days of treatment in which protoporphyrin was measured were the following: subject 1 = 105-238 (7.7 mo), subject 2 = 93-183 (5.9 mo), and subject 3 = 75-208 (6.7 mo). The change in protoporphyrin concentration in either erythrocytes or plasma, assayed by a mixed effects regression method, was not statistically significant.

protoporphyrin (at the Porphyria Laboratory, Galveston, TX).

While receiving colestipol, all subjects experienced an increase in time to photosensitivity onset (Table I). Subject 1 developed photosensitivity after a prolonged delay, and subject 3 lost all photosensitivity. The onset of photosensitivity in subject 2, although prolonged, occurred more rapidly. Despite this variation in prolongation, the colestipol effect was statistically significant. Photosensitivity included itching and burning of the upper extremities, and less frequently pain or erythema of the face and neck. Liver chemistries were repeatedly normal for all subjects. The erythrocyte and plasma protoporphyrin concentrations for each subject (Table II) did not change significantly.

This proof-of-concept study of 3 subjects with EPP has demonstrated that colestipol, a resin that adsorbs and excretes intestinal protoporphyrin, ameliorates the cutaneous complications. The improvement in photosensitivity was greatest in subjects 1 and 3 (Table I). The prolonged onset of photosensitivity was also documented in subject 2, but the shorter prolongation time may result from greater photosensitivity. This effect is similar to the effectiveness of both the new medication afamelanotide,¹ and several other sorbents, at very high dosages. Because we studied only 3 subjects, however, the power is relatively low, and we cannot draw an absolute conclusion regarding protoporphyrin effectiveness.

Although colestipol improves sun sensitivity and probably binds protoporphyrin irreversibly in the ileum,² the mechanism generating skin improvement is not understood. We hypothesize that the sorbent-based irreversible intestinal binding of protoporphyrin and inhibition of enterohepatic reuptake may remove significant quantities of protoporphyrin from skin (thus ameliorating the photosensitivity) and possibly the liver. Although

these likely decreases in cutaneous protoporphyrin did not affect the erythrocyte protoporphyrin (Table II), a longer period of therapy may reduce blood and liver protoporphyrin, thus reducing the liability for liver disease.⁵ We thus urge the conduct of a larger and longer, controlled colestipol trial in subjects with EPP.

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